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FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Mini Arcade-game-Suite

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CERTIFICATE

This is to certify that the project based laboratory report entitled “SMART ACADEMIC PERFORMANCE & ANALYTICS SYSTEM” submitted by **A THOTLA RAM CHARAN, YELLANKI CHAREESHMA, PITHANI CHARAN TEJA , RAKOTI KARTHIK** bearing Regd. No.2500032466, 2500032481, 2500032491, 2500032496 to the **Department of COMPUTER SCIENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in “Python Programming (25SC2007)” course in I B Tech II Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

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ABSTRACT

The Smart Academic Performance & Analytics System (SAPAS) is a comprehensive Python-based platform designed to modernize and automate the management of student academic data. Educational institutions traditionally rely on manual methods such as registers, spreadsheets, and teacher notes, which often result in poor data visibility, lack of insights, and inefficient decision-making. SAPAS addresses these challenges by integrating student information, academic records, attendance tracking, and performance monitoring into a unified digital system, enabling efficient data handling and streamlined academic operations.

The system is structured into multiple intelligent modules that collectively manage core academic functions. It includes student information management, subject-wise score tracking, and attendance processing, along with advanced features like academic log analysis and basic natural language processing for teacher remarks. Additionally, SAPAS incorporates statistical and predictive analytics using data analysis techniques to compute averages, identify performance trends, and forecast potential academic risks. These features help educators quickly identify strengths, weaknesses, and overall student progress.

Furthermore, SAPAS enhances decision-making through effective data storage, retrieval, and visualization mechanisms. Using JSON and CSV for structured data storage and Matplotlib for graphical representation, the system generates insightful charts such as score trends, attendance distributions, and subject comparisons. By automating report generation and providing real-time academic insights, SAPAS serves as a powerful tool for teachers, administrators, and institutions, ultimately improving student performance evaluation and supporting data-driven educational strategies.

Key Words: Smart academic performance & analytics system, Python Programming, Academic Analytics Student Management ,Attendance Tracking ,Performance Analysis ,Predictive Analytics ,Data Visualization ,Python System , Report Automation

INDEX

S.NO	TITLE	PAGE NO
1	Introduction	6
2	Aim of the Project	7
2.1	Advantages & Disadvantages	8
2.2	Future Implementation	8
3	Software & Hardware Details	9
4	Data Flow Diagram	10
5	Algorithm for each module	11
6	Implementation	12-14
7	Integration and System Testing	15-17
8	Conclusion	18

INTRODUCTION

The Smart Academic Performance & Analytics System (SAPAS) is a Python-based application developed to simplify and enhance the management of student academic data in educational institutions. Traditional methods such as registers and spreadsheets often lead to inefficiencies, errors, and lack of meaningful insights. SAPAS overcomes these limitations by integrating student information, subject-wise marks, attendance records, and teacher feedback into a centralized and well-structured system, ensuring better data organization and accessibility.

The system not only automates routine academic tasks like record management, score tracking, and attendance calculation but also provides advanced features such as performance analysis, trend detection, and basic predictive insights. With the help of data visualization tools and structured storage using JSON and CSV formats, SAPAS enables educators to generate reports, monitor student progress, and make data-driven decisions effectively, ultimately improving the overall quality of academic management.

- Python-based academic management system
- Replaces manual record-keeping
- Manages student data, marks, and attendance
- Automates academic tasks
- Provides performance analysis
- Generates reports and visual insights

AIM

The aim of the Smart Academic Performance & Analytics System (SAPAS) is to develop an efficient and intelligent platform for managing and analyzing student academic data. The project focuses on automating essential academic tasks such as record management, marks entry, and attendance tracking while ensuring accuracy and easy accessibility. It also aims to provide meaningful insights through performance analysis, visualization, and report generation, enabling teachers and administrators to make informed, data-driven decisions and improve overall student outcomes.

- To develop a smart system for managing student academic data efficiently
- To automate tasks like marks, attendance, and record management
- To analyze student performance and identify strengths and weaknesses
- To provide data visualization and easy report generation
- To support better academic decision-making using analytics

ADVANTAGES

Advantages of the Project:

- Reduces manual work and saves time
- Improves accuracy in academic records
- Centralized student data management
- Easy tracking of marks and attendance
- Provides clear performance analysis
- Visual representation using charts and graphs
- Supports better decision-making
- User-friendly and efficient system

DISADVANTAGES

Disadvantages of the Project:

- Requires basic technical knowledge to operate
- Initial setup and development time is high
- Depends on system/computer availability
- Data loss risk if not properly backed up
- May need training for teachers/users
- Performance may slow with large data sets

FUTURE IMPLEMENTATION

SAPAS can be enhanced by adding AI/ML for better predictions, cloud storage for scalability, and mobile app support for easy access. Future improvements may also include real-time notifications, advanced dashboards, better security, and integration with learning platforms.

SYSTEM REQUIREMENTS

➤ **SOFTWARE REQUIREMENTS:**

The major software requirements of the project are as follows:

Language: Python

Operating System: Windows 10 or above

Software Tools: Python IDLE / PyCharm / VS Code

Technical Requirements: Monitor, CPU, Keyboard

➤ **HARDWARE REQUIREMENTS:**

The hardware requirements needed to run the software are as follows:

RAM: 4 GB or above

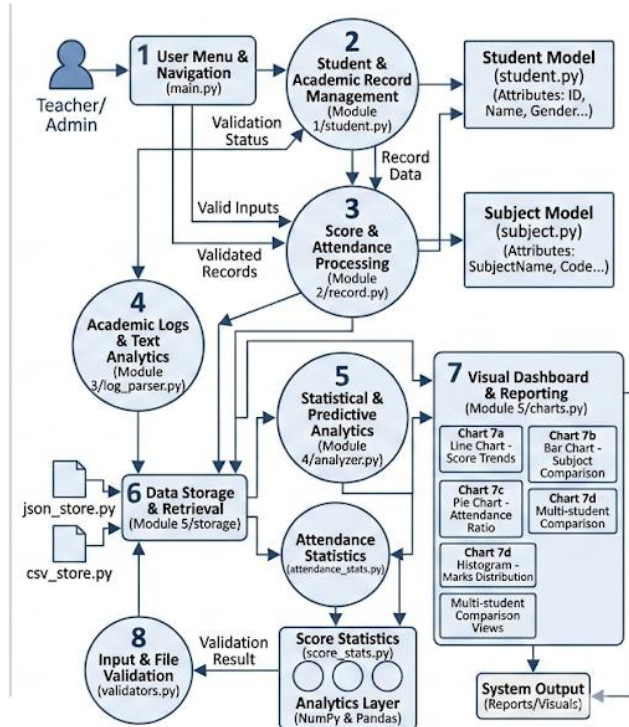
Processor: Intel i3 or higher

DATA FLOW DIAGRAM

Level 0: Context Diagram



Level 1: Detailed DFD



ALGORITHM

->Main Program Algorithm

Step 1: Start

Step 2: Load existing student data from JSON/CSV files

Step 3: Display main menu options

Step 4: Select operation (Add / Update / Delete / View student)

Step 5: If adding student → input and store student details

Step 6: Enter or update subject-wise marks

Step 7: Record attendance data

Step 8: Validate inputs (marks range, student existence, etc.)

Step 9: Store updated data into files

Step10: Perform analysis (average, trends, weak students)

Step11: Generate visual charts (graphs)

Step12: Display results and reports

Step13: Repeat operations until user exits

Step14: Save all data and exit system

IMPLEMENTATION

```
class Student:
    def __init__(self,
roll_no, name, marks):
        self.roll_no =
roll_no
        self.name = name
        self.marks = marks
# dictionary {subject:
marks}

    def total(self):
        return
sum(self.marks.values()
)

    def average(self):
        return self.total() /
len(self.marks)

    def grade(self):
        avg =
self.average()
        if avg >= 90:
            return 'A+'
        elif avg >= 75:
            return 'A'
        elif avg >= 60:
            return 'B'
        elif avg >= 50:
            return 'C'
        else:
            return 'F'

class AcademicSystem:
    def __init__(self):
        self.students = []

    def add_student(self):
        roll = input("Enter
Roll No: ")
        name =
input("Enter Name: ")

        subjects = ["Math",
"Science", "English"]
        marks = {}
```

```

        for sub in subjects:
            marks[sub] =
float(input(f"Enter
marks for {sub}: "))

```

```

        student =
Student(roll, name,
marks)

```

```

self.students.append(stu
dent)
        print("Student
Added
Successfully!\n")

```

```

        def
display_students(self):
            for s in
self.students:
                print("\n-----
-----")
                print(f"Roll No:
{s.roll_no}")
                print(f"Name:
{s.name}")
                print(f"Marks:
{s.marks}")
                print(f"Total:
{s.total()}")
                print(f"Average:
{s.average():.2f}")
                print(f"Grade:
{s.grade()}")

```

```

        def
class_average(self):
            if not self.students:
                print("No data
available!")
            return

```

```

        total_avg =
sum(s.average() for s in
self.students)
        print(f"\nClass
Average: {total_avg /
len(self.students):.2f}")

```

```

        def topper(self):

```

```

        if not self.students:
            print("No data
available!")
            return

```

```

        top =
max(self.students,
key=lambda s: s.total())
        print("\nTopper
Details:")
        print(f"Name:
{top.name}")
        print(f"Roll No:
{top.roll_no}")
        print(f"Total
Marks: {top.total()}")

```

```

def
subject_analysis(self):
    if not self.students:
        print("No data
available!")
        return

```

```

        subjects =
self.students[0].marks.k
eys()

```

```

        print("\nSubject-
wise Average:")
        for sub in subjects:
            avg =
sum(s.marks[sub] for s
in self.students) /
len(self.students)
            print(f"{sub}:
{avg:.2f}")

```

```

def main():
    system =
AcademicSystem()

```

```

    while True:
        print("\n=====
SMART ACADEMIC
SYSTEM =====")
        print("1. Add
Student")
        print("2. Display

```

```

Students")
    print("3. Class
Average")
    print("4. Topper")
    print("5. Subject
Analysis")
    print("6. Exit")

    choice =
input("Enter choice: ")

    if choice == '1':

system.add_student()
    elif choice == '2':

system.display_students
()
    elif choice == '3':

system.class_average()
    elif choice == '4':
        system.topper()
    elif choice == '5':

system.subject_analysis
()
    elif choice == '6':

print("Exiting...")
    break
    else:
        print(" Invalid
choice!")

if __name__ ==
 "__main__":
    main()

```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
"C:\Users\Administrator\EXAM things\.venv\Scripts\python.exe" "C:\Users\Administrator\EXAM things\python code\cherry.py"
```

```
===== SMART ACADEMIC SYSTEM =====
```

1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit

Enter choice: *1*

Enter Roll No: *23001*

Enter Name: *sita*

Enter marks for Math: *80*

Enter marks for Science: *70*

Enter marks for English: *88*

Student Added Successfully!

```
===== SMART ACADEMIC SYSTEM =====
```

1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit

Enter choice: *1*

Enter Roll No: *23003*

Enter Name: *vishnu*

Enter marks for Math: *90*

Enter marks for Science: *80*

Enter marks for English: *88*

Student Added Successfully!

```
===== SMART ACADEMIC SYSTEM =====
```

1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit

Enter choice: 2

```
-----  
Roll No: 23001
```

Name: sita

Marks: {'Math': 80.0, 'Science': 70.0, 'English': 88.0}

Total: 238.0

Average: 79.33

Grade: A

```
-----  
Roll No: 23003
```

Name: vishnu

Marks: {'Math': 90.0, 'Science': 80.0, 'English': 88.0}

Total: 258.0

Average: 86.00

Grade: A

```
===== SMART ACADEMIC SYSTEM =====
```

1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit

Enter choice: 3

Class Average: 82.67

```
===== SMART ACADEMIC SYSTEM =====
```

1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit

Enter choice: 4

Topper Details:

Name: vishnu

```
5. Subject Analysis
6. Exit
Enter choice: 4
```

```
Topper Details:
Name: vishnu
Roll No: 23003
Total Marks: 258.0
```

```
===== SMART ACADEMIC SYSTEM =====
```

```
1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit
Enter choice: 5
```

```
Subject-wise Average:
Math: 85.00
Science: 75.00
English: 88.00
```

```
===== SMART ACADEMIC SYSTEM =====
```

```
1. Add Student
2. Display Students
3. Class Average
4. Topper
5. Subject Analysis
6. Exit
```

```
Enter choice: 6
```

```
Exiting...
```

```
Process finished with exit code 0
```

CONCLUSION

The Smart Academic Performance & Analytics System (SAPAS) provides an efficient and modern approach to managing student academic data by replacing manual methods with a digital solution. It integrates student records, marks, and attendance while offering performance analysis, visualization, and automated reporting. This system improves accuracy, saves time, and supports data-driven decision-making, ultimately helping educators enhance student performance and overall academic management.